

Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher

% Facts

sentence(the_cat_sits).

sentence(the_dog_runs).

% Query

% ?- sentence(X).

Output:

```
11 ?- consult(['c:/Users/Lenovo/Downloads/exp7.1.pl']).
true.

12 ?- sentence(X).
X = the_cat_sits .
```

Exercise 2: Compound Term Unification

% Facts

likes(john, food(pizza, spicy)).

likes(mary, food(pizza, sweet)).

% Query

% ?- likes(john, food(pizza, T)).

Output:

```
13 ?- consult(['c:/Users/Lenovo/Downloads/exp7.2.pl']).
true.

14 ?- likes(john, food(pizza, T)).
T = spicy .
```

Exercise 3: Conflict Resolution

% Facts

```
rule1(climate_change).
```

```
rule2(climate_change).
```

% Rule

```
conflict(X) :- rule1(X), rule2(X).
```

% Query

```
% ?- conflict(climate_change).
```

Output:

```
15 ?- consult(['c:/Users/Lenovo/Downloads/exp7.3.pl']).  
true.  
  
16 ?- conflict(climate_change).  
true.
```

Result

Thus, the Unification and Resolution exercises were successfully executed in PROLOG, and the expected results were obtained.